



Series — DC

279kWh DC Battery Cabinet

Sodium-Ion NFPP technology engineered for critical infrastructure resiliency and demanding commercial deployments — with bankable economics.

279kWh

NAMEPLATE ENERGY

229kWh

USABLE (85% DOD)

140kW

MAX RATED POWER

14%

LOWER TCOO VS. LFP

634-994V

DC VOLTAGE RANGE

2.2-10^{hr}

DURATION RANGE

-40/+55°C

OPERATING TEMP

IP54

ENCLOSURE RATING



ENERGY STORAGE YOU CAN BANK ON

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UL 1973

UL 9540

UL 9540A

UN 38.3

IP54

IEC 62619

PRODUCT OVERVIEW

279kWh DC Battery Cabinet

- Sodium-Ion NFPP
- Air-Cooled
- 2.2–10 Hour Duration



Built for critical infrastructure resiliency and demanding use cases, the Series DC 279kWh Battery Cabinet prioritizes compliance, field-validated performance, and bankable warranties — delivering a **14% lifetime TCoO improvement** over liquid-cooled lithium-ion systems.

Includes an outdoor, IP54-rated enclosure containing:

- Sodium-ion NFPP battery modules
- Air-cooled thermal management
- Fire suppression & gas detection
- Battery Management System (BMS)
- Auxiliary safety systems
- Explosion venting system

KEY BENEFITS



Safety-First Design

Sodium-ion chemistry eliminates thermal runaway risk from the chemistry up — enabling safer siting and simpler AHJ approvals.

- ✓ Lower lifetime O&M costs
- ✓ Air-shippable — faster deployment
- ✓ Reduced thermal propagation risk
- ✓ Clearer AHJ approvals



Extreme Performance

Operates where traditional batteries fail — from arctic cold to desert heat — without derating or HVAC infrastructure.

- ✓ -40°C to +55°C operation
- ✓ 20-year projected battery life
- ✓ No HVAC required
- ✓ Fast charging capability



Supply Chain Resilience

Real-world validation, not laboratory promises. Clear domestic manufacturing pathway with no tariff exposure.

- ✓ DOD / Federal compliance path
- ✓ No tariff exposure
- ✓ Domestic manufacturing pathway
- ✓ Supply chain diversification

WHERE THE SERIES DC PERFORMS



MANUFACTURING / COLD STORAGE / DISTRIBUTION

Peak Demand Management

Reduce demand charges by 30–50% at manufacturing facilities, cold storage warehouses, distribution centers, and agriculture operations while maintaining full capacity in extreme temperatures.



DATA CENTERS / HEALTHCARE / TELECOM

Critical Infrastructure Backup

Guaranteed uptime for data centers, hospitals, emergency services, and telecommunications — with air-cooled simplicity and no fire suppression complexity.



FORKLIFT / WAREHOUSE / PORT / AIRPORT

Material Handling & Fleet

Fast charging and extreme temp tolerance for 24/7 operations — forklift fleets, warehouse automation, port equipment, and airport ground support.



SOLAR / WIND / CAMPUS / COMMUNITY

Renewable Energy Integration

Long-duration storage with 20-year design life for bankable projects — solar + storage microgrids, wind energy smoothing, campus energy systems, and community solar projects.

SERIES DC — CABINET



NAMEPLATE ENERGY

279kWh

229 kWh usable at 85% DOD

MAX RATED POWER

140kW

125 kW guaranteed constant

DC VOLTAGE RANGE

634–994v

Flexible inverter compatibility

DURATION RANGE

2.2–10hr

Peak shaving to long backup

OPERATING TEMPERATURE

–40/+55°C

No HVAC — any climate

ENCLOSURE RATING

IP54

Outdoor-rated direct install

PARAMETER	VALUE / DESCRIPTION	UNIT
NAMEPLATE ENERGY CAPACITY	279	kWh
USABLE ENERGY (85% DOD)	229	kWh
MAX RATED POWER	140	kW
GUARANTEED CONSTANT POWER	125	kW
DC VOLTAGE RANGE	634 – 994	VDC
DURATION	2.2 – 10	Hours
CELL TYPE	Sodium-Ion NFPP (Non-Flammable Polymer Pouch)	—
OPERATING TEMP (DISCHARGE)	–40°C to +55°C	—
DIMENSIONS (L × W × H)	2.0 × 1.5 × 2.3	Meters
ENCLOSURE (IP RATING)	Outdoor Cabinet — IP54	—
THERMAL MANAGEMENT	Air-Cooled (No HVAC Required)	—
BMS COMMUNICATION	US BMS, Modbus TCP	—
AUXILIARY SYSTEMS	Fire Suppression, Gas Detection, Venting	—

CERTIFICATIONS

UL 1973

UL 9540

UN 38.3

Cell & Module Level

UL 9540A TESTING

Cell: Tested & Passed

Module/System: In Progress

WARRANTY

Cell: 3 Years

System: 5 Years

Extended warranty available

FACTORY ACCEPTANCE TEST (FAT) SCOPE

- ✓ Visual, Mechanical & Torque Inspection
- ✓ 3 × 0.125C Capacity/Energy Verification (85% DOD)
- ✓ Protections & Isolation Tests
- ✓ Baseline OCV Measurement
- ✓ Functional & Conditioning Test
- ✓ 10 × 0.5C Stability Cycles with Power Checks

Sodium-Ion NFPP vs. Lead-Acid

• Performance • Safety • Total Cost of Ownership

WHY SODIUM-ION NFPP OUTPERFORMS LEGACY LEAD-ACID TECHNOLOGY

A Clear Advantage Across Every Performance Dimension

Coulomb Technology's Sodium-Ion NFPP chemistry delivers superior cycle life, wider operating temperature range, zero maintenance, and a fundamentally safer chemistry — making it the ideal replacement for lead-acid in critical commercial and industrial applications.

FEATURE	SODIUM-ION NFPP	LEAD-ACID
Cycle Life	3,000+ cycles @ 80% DoD	300–500 cycles @ 50% DoD
Weight (100Ah)	13 kg (28.7 lb)	~22 kg (48.5 lb)
Usable Capacity	80–100% DoD	50% DoD (to preserve life)
Discharge Temp Range	–40°C to +60°C	–20°C to +50°C
Charge Temp Range	–10°C to +45°C	0°C to +40°C
Maintenance	Zero maintenance	Regular watering & equalization
Self-Discharge	<3% per month	5–15% per month
Charging Efficiency	~95%	~75%
Thermal Runaway Risk	Zero risk	Low risk
Toxic Gas Emissions	None	Hydrogen gas during charging
Memory Effect	None	Sulfation if undercharged
Lifespan	10+ years	3–5 years
Environmental Impact	Non-toxic, recyclable	Lead & acid hazardous

10×

Longer cycle life
vs. lead-acid

40%

Lighter weight
per 100Ah

Zero

Maintenance
required

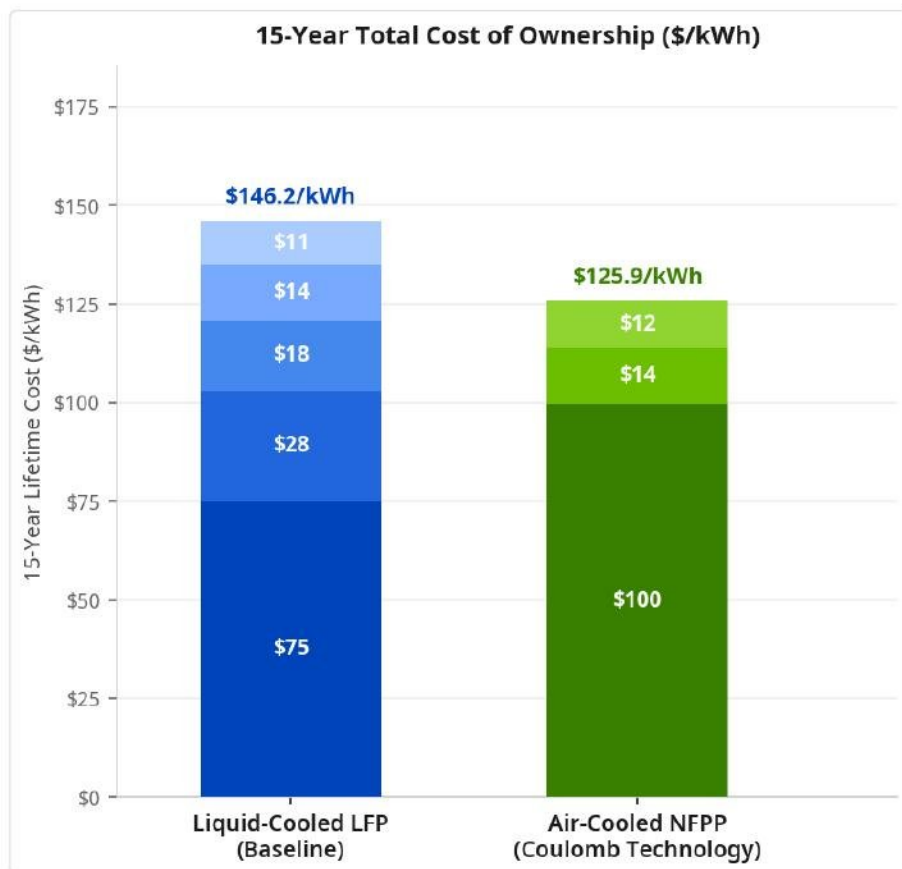
95%

Round-trip
efficiency

SYSTEM ADVANTAGE OVER LFP-BASED SYSTEMS

14% Lower Total Cost of Ownership Over the Project Lifetime

Our sodium-ion technology delivers measurable savings through eliminated HVAC requirements, reduced degradation, and lower maintenance, insurance, and permitting costs — validated by real-world deployments, not laboratory projections.



MODEL ASSUMPTIONS

NFPP installed price delta	\$16/kWh
Cycles per year	365
HVAC load delta	90%
Insurance cost delta	20%
Augmentation delta	20%
O&M cost delta	30%
Cell derating / cold temps	75%
Discount rate / NPV calc	8%

COST COMPONENTS

LFP — CapEx	NFPP — CapEx
LFP — HVAC Load	NFPP — O&M
LFP — O&M	NFPP — Insurance
LFP — Insurance	LFP — Augmentation

14% TCcO Savings — Air-cooled NFPP (\$125.9/kWh) vs. Liquid-cooled LFP (\$146.2/kWh) over a 15-year project lifetime.

\$20.3/kWh — Lifetime savings per kWh over project lifetime

20yr — Projected battery life, sodium-ion NFPP chemistry

RESEARCH & DEVELOPMENT PARTNERS

